

Master/
Orchestrator

SU#
(Gen 0)

SU#
(Gen 1)

Gauss-Seidel Co-Simulation: One Step $t \rightarrow t+dt$

For each generation sequentially

Generation 0: Process SU#

1. Set inputs from connections

Use outputs from
previous time step

inputs set

alt

[SU# in algebraic loop]

2. Solve algebraic loop (IJCSA)

Interface-Newton iteration
to resolve coupling

converged

3. Execute time step $\text{do_step}(t, dt)$

Integrate $t \rightarrow t+dt$

Update outputs at $t+dt$

$Y\#(t+dt)$ available

Generation 1: Process SU#

1. Set inputs from connections

Use **fresh** $Y\#(t+dt)$
from SU#

inputs set

alt

[SU# in algebraic loop]

2. Solve algebraic loop (IJCSA)

converged

3. Execute time step $\text{do_step}(t, dt)$

Integrate $t \rightarrow t+dt$

Update outputs at $t+dt$

$Y\#(t+dt)$ available

All generations complete
 $t \leftarrow t + dt$

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